Simoa blood biomarkers tested the Quanterix Neurology 4-Plex E and ptau181V2 assays on Insight46 phase 1 and 2

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Document details

Categories of variables	Simoa blood biomarkers tested the Quanterix Neurology 4-Plex E and ptau181V2 assays on phase 1 and 2 plasma samples, and phase 2 CSF samples from the Insight 46 study
Summary of work undertaken	Source data transferred to Stata file Age variables derived Missing data – labelled as -99. Methods section below
Names of source variables	NFL_N4PE_plasma_p1, GFAP_N4PE_plasma_p1, Ab40_N4PE_plasma_p1, Ab42_N4PE_plasma_p1, Ptau181v2_plasma_p1, NFL_N4PE_plasma_p2, GFAP_N4PE_plasma_p2, Ab40_N4PE_plasma_p2, Ab42_N4PE_plasma_p2, Ptau181v2_plasma_p2, NFL_N4PE_CSF_p2, GFAP_N4PE_CSF_p2, Ab40_N4PE_CSF_p2, Ab42_N4PE_CSF_p2, Ptau181v2_plasma_p2, visit_date_p1, Date_blood_p2, Date_CSF_p2
Output variables	age_blood_i46p1, n4pe_plsmft_i46p1, ptau181v2_plsmft_p1, ab42_n4pe_plsm_i46p1, ab40_n4pe_plsm_i46p1, nfl_n4pe_plsm_i46p1, gfap_n4pe_plsm_i46p1, ptau181v2_plsm_i46p1

Methods

General note

All measurements were conducted by the same researcher (Dr Ashvini Keshavan) on the same two automated HD-X analysers (Quanterix) at the UK DRI laboratory at UCL.

For these blood biomarker variables, the code -99 indicates that no aliquot was available for testing so no quantification was performed.

It is recommended that prior to making a request for any of these variables, researchers discuss the plan with Dr Ashvini Keshavan.

Age variable derivation

Each of the three age variables (Age_at_blood_p1, Age_at_blood_p2 and Age_at_CSF_p2) was derived by subtracting the participant date of birth from the date of sampling and then dividing the resulting value in days by 365.25 to obtain the age at sampling in years:

Age_at_blood_p1 = (visit_date_p1 – dob)/365.25 Age_at_blood_p2 = (Date_blood_p2 – dob)/365.25 Age_at_CSF_p2 = (Date_CSF_p2 – dob)/365.25

EDTA plasma – samples from phases 1 and 2

EDTA plasma was obtained from non-fasted participants at Insight 46 Phase 1 and phase 2 as previously described (Keshavan et al. Brain 2021; 144: 434-449).

Prior to analysis, a single 500 µL aliquot of plasma for each individual was thawed directly to room temperature over 1 hour and vortexed for 2 seconds to ensure thorough mixing. 300 µL was pipetted into a 1.5 ml low-bind polypropylene centrifuge tube and centrifuged at 10 000 g for 10 minutes, as per the kit manufacturer's recommendation; the remaining 200 µL in the original cryovial was replaced into -80 ºC. After the 300 µL was centrifuged, 130 µL of the supernatant was pipetted onto a plate for analysis using the Neuro 4-Plex E assay (Quanterix) on a Simoa HD-X platform. Another 130 µL of the supernatant was pipetted onto a separate plate for analysis using the ptau181V2 assay (Quanterix) on a different Simoa HD-X platform.

Concentrations of neurofilament light-chain (NfL), glial fibrillary acidic protein (GFAP), amyloid beta 40 (Aβ40) and amyloid beta 42 (Aβ42), were quantified by the Neuro 4-Plex E multiplex assay, and concentration of tau phosphorylated at serine 181 (p-tau181) were quantified by the ptau181V2 assay, from Phase 1 and 2 plasma samples in singlicate, according to the manufacturer's instructions, with 4x in-line dilution.

CSF – samples from phase 2

CSF was obtained from non-fasted participants as previously described (Keshavan et al. Alz Dem (Amst.) 2021; 13: e12131) and stored at –80 C within 60 min of sampling.

Prior to analysis, a single 500 μL aliquot of CSF for each individual was thawed directly to room temperature over 1 hour and vortexed for 2 seconds to ensure thorough mixing. For N4PE with 400x at bench dilution, 40 μL of CSF was pipetted into a 1.5 ml low-bind polypropylene centrifuge tube containing 360 μL of N4PE sample diluent and vortexed. 250 μL of the diluted sample was pipetted onto a plate for analysis using the Neuro 4-Plex E assay (Quanterix) on a Simoa HD-X platform. For ptau181V2 with 10x at bench dilution, 30 μL of CSF was pipetted into a 1.5 ml low-bind polypropylene centrifuge tube containing 270 μL of ptau181V2 sample diluent and vortexed. 200 μL of the diluted sample was pipetted onto a plate for analysis using the ptau181V2 assay (Quanterix) on a different Simoa HD-X platform. Where possible, paired plasma and CSF from the same individuals were analysed on the same plates.

Repeat measurements and freeze-thaw cycles

If the initial quantification failed for a plasma sample, the same aliquot (with remaining approximately 200 μL having undergone one prior freeze-thaw cycle) was thawed again over 1 hour to room temperature and centrifuged in the original cryovial at 10 000 g for 10 minutes, then 130 μL of the supernatant was pipetted onto a plate for analysis using the N4PE or ptau181v2 as above.

If the initial quantification of a CSF sample failed, the same original CSF aliquot (with remaining approximately 430 μL of CSF having undergone one previous freeze-thaw cycle) was thawed again over 1 hour to room temperature, vortexed for 2 seconds and the required volume for the N4PE or ptau181V2 assay pipetted out for on-bench dilution as per the procedure described above.

By this method, 12 plasma samples from phase 1, 12 plasma samples from phase 2 and 2 CSF samples from phase 2 analysed using the N4PE assay underwent two freeze-thaw cycles. 18 plasma samples from phase 1, 3 plasma samples from phase 2 and 1 CSF sample from phase 2 analysed using the ptau181V2 assay underwent two freeze-thaw cycles. This procedure was deemed to be acceptable based on the evidence that up to four freeze-thaw cycles do not affect either the N4PE-measured biomarker concentrations or plasma p-tau181 concentrations on a different plasma p-tau181 assay on the MSD platform (Verberk et al. Alz Dem 2021;18(8):1484–1497) and up to three freeze-thaw cycles do not affect measured CSF p-tau181 or Aβ42 concentrations on a different assay platform (Innotest ELISA; Zimmermann et al. JAD 2011; 25: 739-745).

Control variation

For each assay, all samples were tested using the same batch of reagents. Every plate was analysed according to its own calibrators (provided with the kits). Five pooled plasma controls (made from remnants of samples provided kindly by Prof Henrik Zetterberg at the University of Gothenburg and pooled according to their plasma NfL ranges) and two run validation controls provided with the kits were also assayed with every run. Plate acceptance criteria were set as at least 4 of 7 controls being within 1 standard deviation of the mean value, and all 14 plates (across 9 runs) met these criteria. Control mean values and coefficients of variation (CV) were as follows:

Control	Mean NfL (pg/ml)	CV (%)	Mean GFAP (pg/ml)	CV (%)	Mean Ab40 (pg/ml)	CV (%)	Mean Ab42 (pg/ml)	CV (%)	Mean ptau181v2 (pg/ml)	CV (%)
Pooled plasma 1	14.6	21.9	107	32.0	4.8	23.4	67.6	37.4	2.1	10.9

Pooled plasma 2	29.0	6.5	102	10.9	3.8	9.4	51.7	8.5	2.7	10.1
Pooled plasma 3	49.1	9.3	190	15.3	3.7	16.8	60.8	10.6	2.6	7.0
Pooled plasma 4	69.1	7.5	157	12.1	3.8	10.2	48.4	10.4	2.5	13.8
Pooled plasma 5	1054	27.6	229	12.3	4.8	8.5	71.4	12.4	11.3	7.0
Run validation control 1	23.3	11.8	175	16.6	5.4	23.4	14.6	20.2	3.6	38.2
Run validation control 2	519.4	10.9	3768	16.4	28.2	7.9	99.2	20.1	99.7	77.7

Variables in this document

7 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Biological samples [6037] › Blood biomarkers [60372]		
ab40_n4pe_plsm_i46p1	Plasma Ab40 from Neuro 4-Plex E assay (N4PE), pg/ml - Insight46 phase 1	topsheet field
ab42_n4pe_plsm_i46p1	Plasma Ab42 from Neuro 4-Plex E assay (N4PE), pg/ml - Insight46 phase 1	topsheet field
gfap_n4pe_plsm_i46p1	Plasma glial fibrillary acidic protein (GFAP) concentration from Neuro 4-Plex E assay (N4PE) 2021, pg/ml - Insight46 phase 1	topsheet field
n4pe_plsmft_i46p1	Plasma Neuro 4-Plex E assay (N4PE) freeze-thaw cycles - Insight46 phase 1	topsheet field
nfl_n4pe_plsm_i46p1	Plasma neurofilament light-chain (NFL) concentration from Neuro 4-Plex E assay (N4PE) 2021, pg/ml - Insight46 phase 1	topsheet field
ptau181v2_plsm_i46p1	Plasma ptau181 concentration from ptau181V2 2021 (pg/ml) - Insight46 phase 1	topsheet field
ptau181v2_plsmft_p1	Plasma ptau181V2 freeze-thaw cycles - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan

1 low-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Sociodemographics [515] › Household [5152]		
own	Now I would like to get some general information about your household. Does your household own or rent this accommodation? - at age 53 years	prose scan